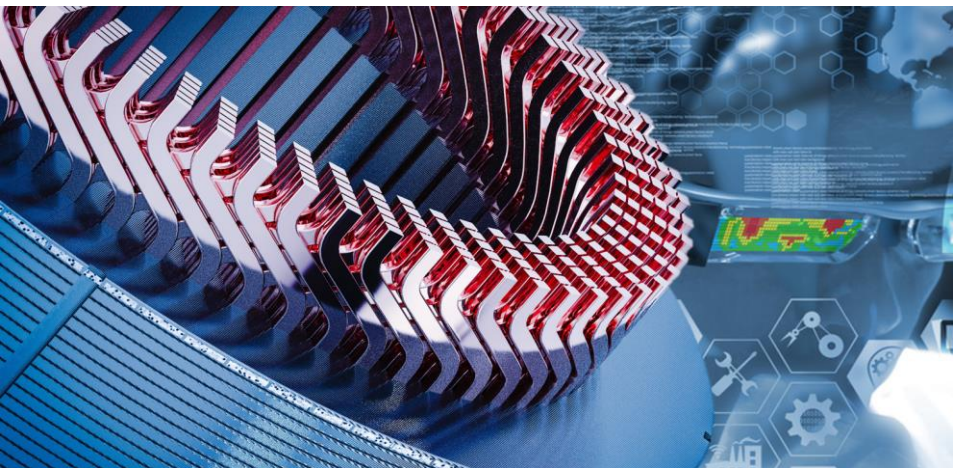




# PRODUCTION PROCESS OF A HAIRPIN STATOR



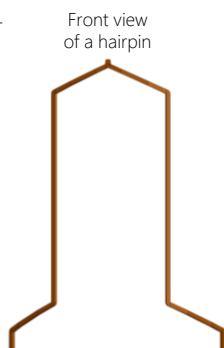
**RWTH**AACHEN  
UNIVERSITY



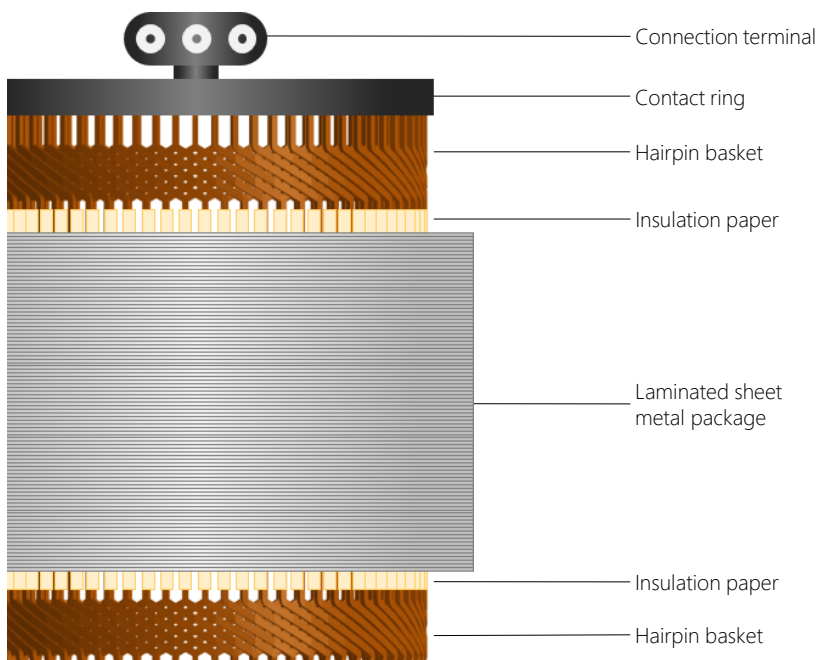
# Hairpin technology

## in contrast to wire winding technology

- The hairpin technology uses massive copper wires (usually flat copper wires) which are inserted in the stator slots of the laminated sheet metal package
- Analogous to conventional stators, the laminated sheet metal package consists of numerous layers of electrical sheets insulated against each other (sheet thickness: 0.1 – 1.0 mm)
- The U-shaped hairpins (see illustration on the right) thus replace the classic winding processes such as needle winding, flyer winding and the pull-in technique
- In contrast to the classic processes, the hairpin technology is a forming based assembly process
- The hairpin technology is already widely used in generators and currently shows a strong trend towards traction motors in the automotive industry
- The hairpin technology is suitable for stator winding of synchronous and asynchronous motors with distributed winding
- The electrical slot filling factor of hairpin stators is very high (approx. 73%) due to the rectangular cross section and the small number of windings and is thus considerably higher than the slot filling factor of conventional stators (45-50%, depending on the insulation thickness)
- In the context of hairpin stator production in the automotive industry, cost efficiency is currently achieved at a level of about 200,000 units



## Structure and components of the hairpin stator



# Potentials

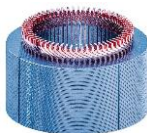
## of the hairpin stator

### Economic efficiency

- High automation capacity
  - High scalability
  - Reduced production costs
- Series production technology

### Process capability

- No stochastic winding
  - Deterministic assembly processes
  - Deterministic welding processes
- Suitable for automobiles



### Winding design

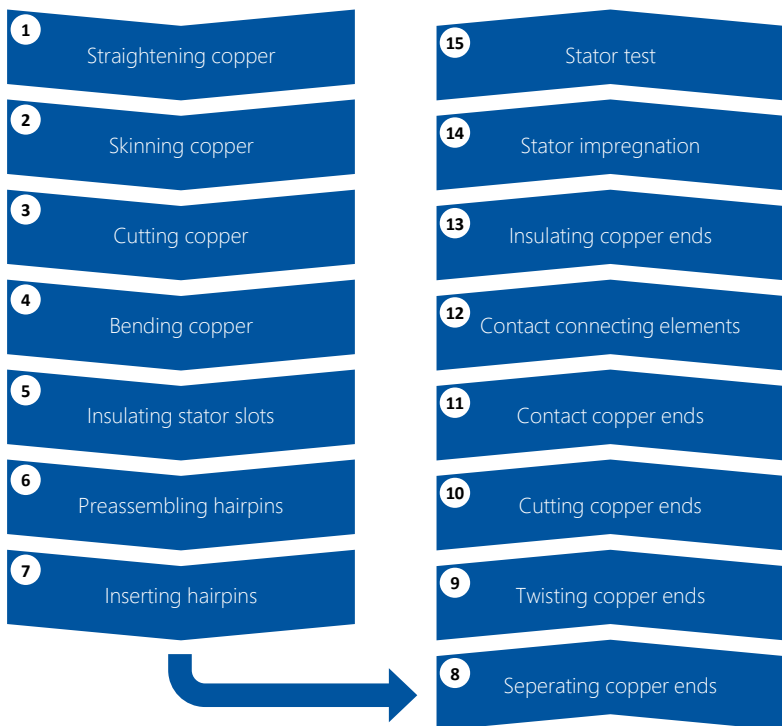
- Large conductor cross section
  - High currents
  - Consideration of the skin effect
  - Limited flexibility
- High torque motors

### Technological maturity

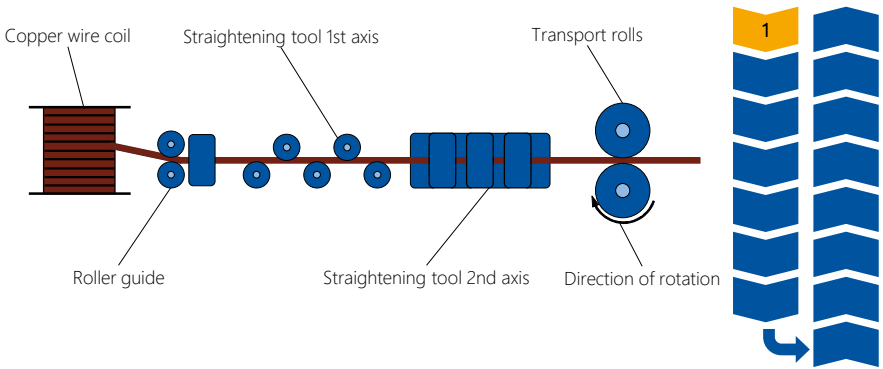
- Industrialization currently in progress
  - High technological potential
  - Production technology as enabler for product development
- High potential for suppliers

# Process chain

## of a hairpin stator production



# Straightening copper



## Process description

- The winding of the copper wire onto the coil causes curvatures in the wire structure, which have to be corrected for the further process steps
- In multi-stage straightening, the wire is straightened in several steps by straightening tools (in two axial directions) to remove the residual curvature as far as possible
- The exact straightening of the rectangular copper wire leads to an increase in positioning accuracy when setting and welding the hairpins in the downstream process steps
- The transport rolls then convey the straightened copper wire to the next process step
- As the straightening process is very sensitive to fluctuations in input parameters, the upstream decoiler process, the wire transport as well as the wire properties have to be controlled

## Further information

### Technology alternatives [extract]

- Stretch straightening
- Stretch bending straightening
- Additional: Inline and offline control of the wire before and after the process

### Quality characteristics [extract]

- Low helicity
- Low (residual) curvature
- Low superelevation
- Form tolerances: changes in cross section
- Low wire damage
- Low impurities

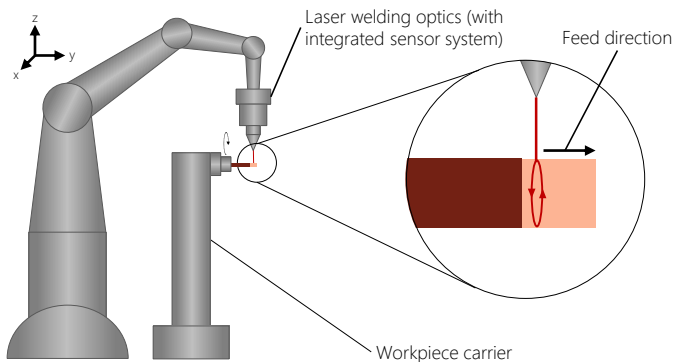
### Quality influences [extract]

- Straightening roll position
- Material springback and yield strength
- Wire feed speed and continuity
- Straightening material handling
- Residual stress state
- Tool rigidity

### Production costs\* [extract]

Machinery/equipment investment: € 0.01 – 0.05 million

# Skinning copper at the copper wire ends



## Process description

- For the subsequent contacting of the copper ends (process step 11), the copper ends have to be selectively freed from their insulating layer
- The area to be stripped is approached several times by the laser (or by several lasers with different wavelengths) in pulsed mode
- First, the insulation "burns" with a visible flame, as the conductor and the insulating material are heated by the laser
- After removing most of the insulation, the laser radiation is absorbed by the base material
- Due to an abrupt thermal heating of the conductor, remaining insulation and combustion residues are "blasted off"

## Further information

### Technology alternatives [extract]

- Mechanical processes
  - Grinding
  - (Ultrasound) scraping
  - Milling
  - Brushing
- Chemical processes
  - N-methyl-2-pyrrolidone (NMP), N-ethyl-2-pyrrolidone (NEP),  $\gamma$ -butyrolactone (GBL)

### Quality characteristics [extract]

- Stripped area: Low insulation residues and residues of scaled material
- No removed or thermally highly stressed conductor material
- Positioning accuracy of the stripping
- High repeatability/process reliability

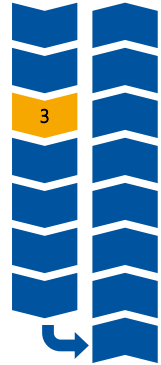
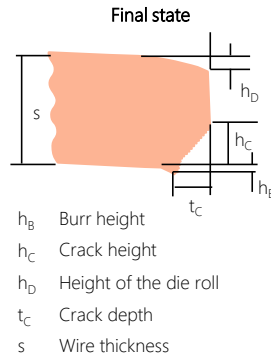
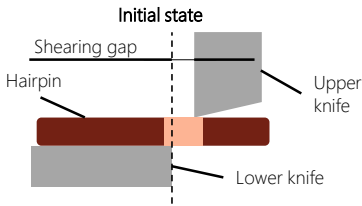
### Quality influences [extract]

- Beam source/wavelength
- Process speed (feed rate)
- Laser power
- Pulse duration
- Welding feed, focusing
- Insulation material

### Production costs\* [extract]

Machinery/equipment investment: € 0.08 – 0.4 million  
(per machine)

# Cutting copper at the copper wire ends



## Process description

- The straightened copper wire is cut to length according to the required hairpin geometry
- The shearing process consists of the following four phases:
  - Elastic deformation
  - Plastic deformation
  - Plastic flow
  - Crack formation
- The shearing process must be adjusted in such a way that a deformation of the wire cross section and a burr formation is avoided
- Optional: Integrated sharpening/chamfering of the copper ends for easier assembly

## Further information

### Technology alternatives [extract]

- Shear cutting
- Knife cutting
- Bite cutting
- Tearing

### Quality characteristics [extract]

- Position tolerance (positioning of the cut)
- Shape tolerance of the chamfer
- Cutting burr
- No damage to the conductor insulation
- No deformation of the wire cross section

### Quality influences [extract]

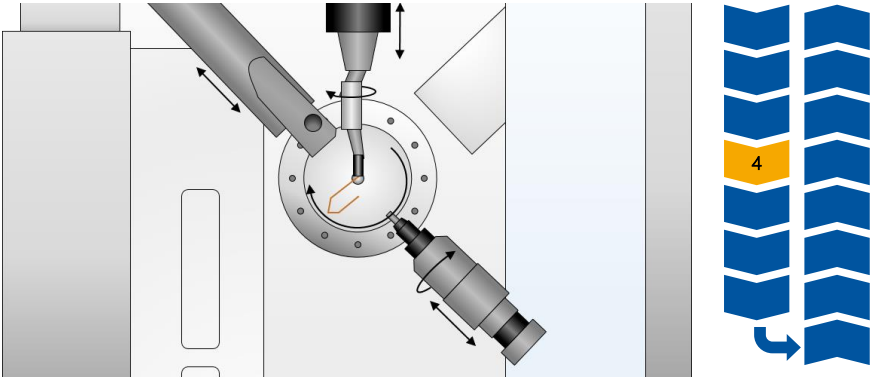
- Material thickness and material type (fluctuating)
- Hold-down force
- Cutting gap in relation to material thickness
- Wear condition of the cutting tools
- Cutting speed

### Production costs\* [extract]

Machinery/equipment investment: € 0.01 – 0.05 million  
(per machine)

# Bending copper

## 2D and 3D bending (CNC bending)



## Process description

- Within the scope of the bending process, a distinction is made between 2D and 3D bending
- The copper wires have to be bent three-dimensionally in order to be able to connect different layers in the stator
- The entire shaping process of the copper wires can be carried out in one or more stages (integrated or specialised production plants)

## Further information

### Technology alternatives [extract]

- CNC bending: Computer-aided control of the bending machines
- Swivel bending: The workpiece is clamped between the upper and lower cheek of the bending tool
- Die bending: The pre-bent workpiece is placed on a die with a V-shaped opening
- Free bending/pulling: The final shape is only controlled by the movement of the tool
- The technology alternatives are illustrated and described on the following page

### Quality characteristics [extract]

- Form, angle and position tolerances (absolute position/rotation, opening angle)
- No insulation damage or change in insulation thickness on the outer radius
- Change in cross-section/residual stress state
- Repeat accuracy (scatter of results)

### Quality influences [extract]

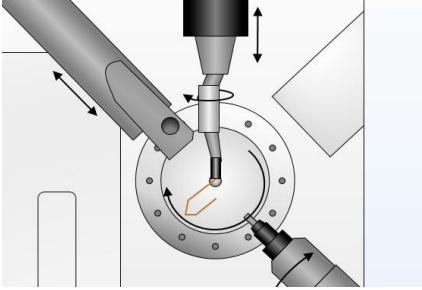
- Forming time, speed and (bending) force
- Punch radius (geometry-dependent)
- Springback compensation
- Material parameters (e.g. modulus of elasticity, yield strength)
- Quality and wear of the bending tool

### Production costs\* [extract]

Machinery/equipment investment: € 0.2 – 1.5 million  
(per machine)

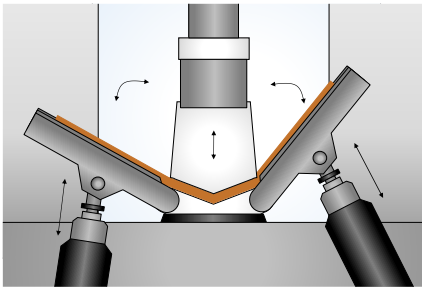
# Bending copper

## Technology alternatives



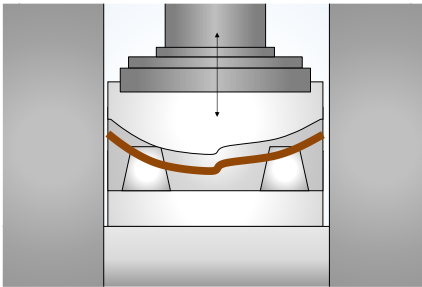
### CNC bending

- Integrated 2D and 3D bending process
- Copper wire is brought directly to the final shape by a single or multiple bending tools
- Highly flexible process for the production of different hairpin geometries on one machine
- Comparatively long cycle time



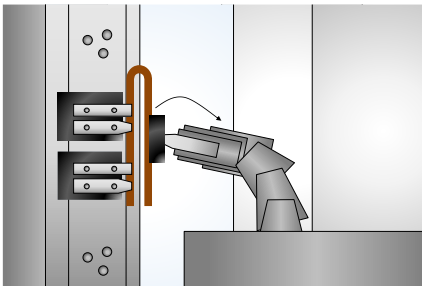
### Swivel bending

- Wire is positioned on the stop system
- Upper bending beam tensions the copper wire against the lower bending beam
- The bending beams give the copper wire the required 2D geometry
- Use in combination with die bending or free bending (3D bending processes)



### Die bending

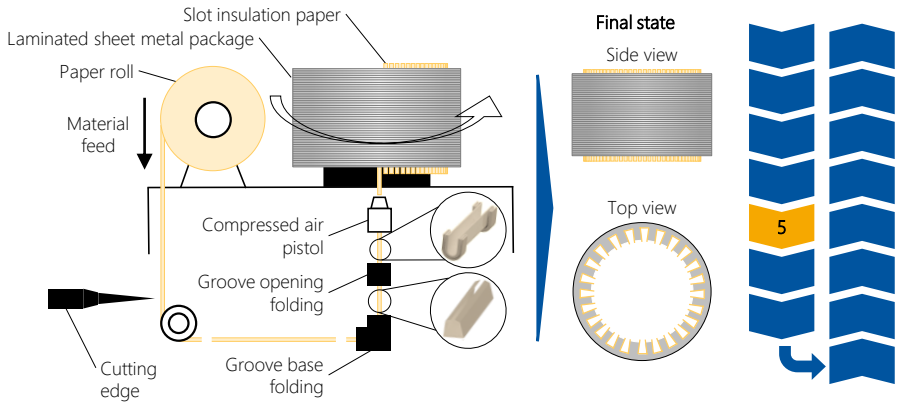
- 2D-bended hairpin is positioned on the die
- By moving the die downwards, the hairpin is given the 3D shape
- High reproducibility with short cycle time
- Use in combination with e.g. swivel bending



### Free bending / pulling

- One leg of the 2D-shaped hairpin is positioned between the upper and lower cheek
- Gripper takes free hairpin leg and pulls the hairpin apart
- Use in combination with e.g. swivel bending

# Insulating stator slots



## Process description

- The stripped hairpins are separated from the ground potential of the laminated sheet metal package by using insulation paper
- Manual or fully automated cutting and folding (groove base folding and groove opening folding) of the insulation paper possible
- A continuous synchronization of the material feed, the timing of insertion or shooting, the forming force and kinematics during the forming and lining of the slots is required
- Optional: Consolidation of the insulation paper in the stator slots and funnel-shaped expansion of the insulation paper overhang for easier assembly
- Manual or automated inspection for damage and freedom from cracks and creases

## Further information

### Technology alternatives [extract]

- Plastic injection moulding: Insulation of the slots with a process similar to injection moulding
- Powder insulation: Application of powder resins or varnishes and subsequent curing
- 3D printing: Layer-by-layer extrusion of high-viscosity pastes and subsequent sintering of the component

### Quality characteristics [extract]

- Deviation in shape and position of the insulation
- No damage to the axial final edge of the stator
- Kink resistance and freedom from cracks/creases
- Air inclusions (between slot insulation and the sheet metal package)
- Defined insulation paper overhang

### Quality influences [extract]

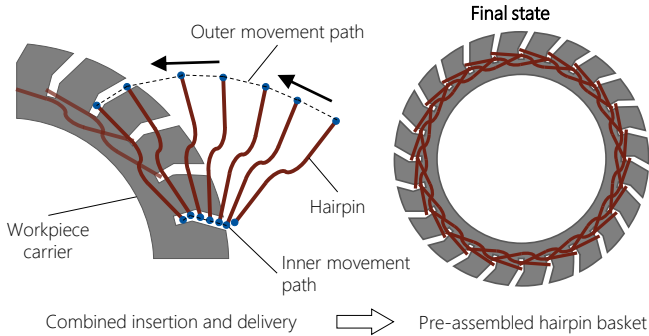
- Stator geometry (slot geometry and length)
- Folding scheme
- Winding process
- Insertion speed
- Forming force

### Production costs\* [extract]

Machinery/equipment investment: € 0.1 – 0.4 million

# Preassembling hairpins

## to a hairpin basket



## Process description

- Undercuts and overlaps of the winding head geometry limit the sequential direct assembly of the hairpins in the laminated sheet metal package and thus require a pre-assembly
- The hairpins are pre-assembled outside the laminated sheet metal package to form one or more hairpin baskets
- The basket assembly takes place in a workpiece carrier or a transfer device
- Pre-assembly strategies such as turning or radial feed are used
- During pre-assembly, the hairpins must be sorted correctly (approx. 3 - 16 types per stator)
- The high dependence of pre-assembly on the winding scheme and hairpin geometry requires complex special machine constructions and limits the flexibility of variants

## Further information

### Technology alternatives [extract]

- Direct assembly
- Continuous hairpin winding
- Integration of bending operations into the pre-assembly operation

### Quality characteristics [extract]

- No damage to the conductor insulation
- No plastic deformation of the hairpins
- Form and positional accuracy of the pre-assembly module

### Quality influences [extract]

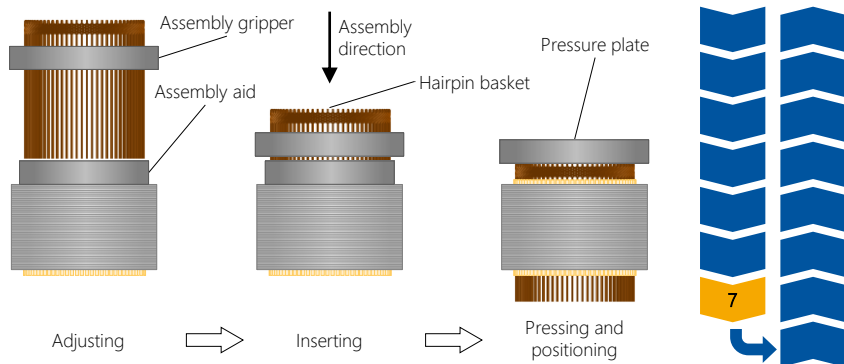
- Handling and clamping forces
- Surface of the parts in contact with the component
- Hairpin geometry deviations
- Undercuts and overlaps

### Production costs\* [extract]

Machinery/equipment investment: € 0.05 – 0.8 million

# Inserting hairpins

Insert hairpins axially in the laminated sheet metal package



## Process description

- The hairpin baskets are moved from the pre-assembly station by multiple handling systems and inserted axially into the laminated sheet metal package (joining gap usually  $< 0.1$  mm)
- The final adjustment of the hairpins is carried out in the handling system or by additional assembly aids
- During the assembly process the slot insulation must be secured against damage and axial shifting
- Drafts on the hairpins and/or fixtures facilitate insertion into the stator slots
- If necessary, special hairpins can be inserted individually into the stator lamination package in order to reduce the wiring effort in the following process steps
- Finally, the hairpins are aligned in their axial end position (e.g. by using a pressure plate)

## Further information

### Technology alternatives [extract]

- Direct assembly
- Continuous hairpin winding

### Quality characteristics [extract]

- No damage to the conductor insulation
- No damage and dislocation of the slot insulation
- No plastic deformation of the hairpins
- Form and position accuracy of the winding head and the free copper ends

### Quality influences [extract]

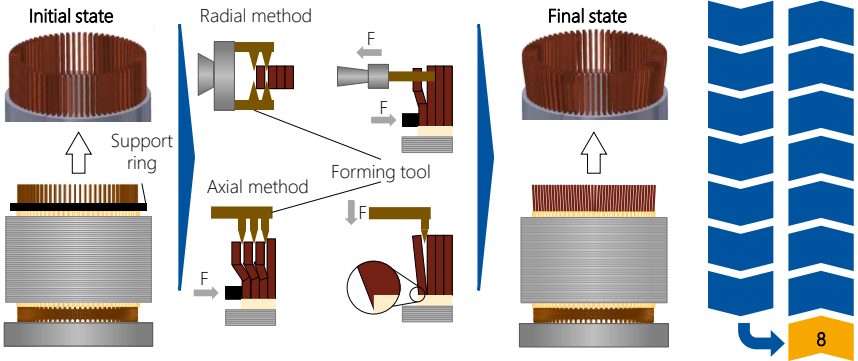
- Slot geometry and joining gap
- Joining direction
- Joining force
- Hairpin geometry deviations
- Handling and clamping forces
- Surface of parts in contact with components

### Production costs\* [extract]

Machinery/equipment investment: € 0.05 – 0.8 million

# Seperating copper ends

## Preparation of the copper ends for twisting



## Process description

- Radial separation of the copper ends for axial access to the twisting tool
- Supporting the copper ends above the slot insulation is necessary to ensure a defined forming process on the twisted side of the stator
- The copper ends are radially shaped by a forming tool to create sufficient distance between them
- The process can take place in slots, layers or in one step, which has an influence on the achievable cycle time and variant flexibility

## Further information

### Technology alternatives [extract]

- Axial, wedge-shaped forming tool for individual or grouped copper ends
- Continuous hairpin winding
- 3D-printed winding head

### Quality characteristics [extract]

- Geometric tolerances of the copper ends
- No damage to the conductor insulation
- No damage to the slot insulation
- Height tolerance

### Quality influences [extract]

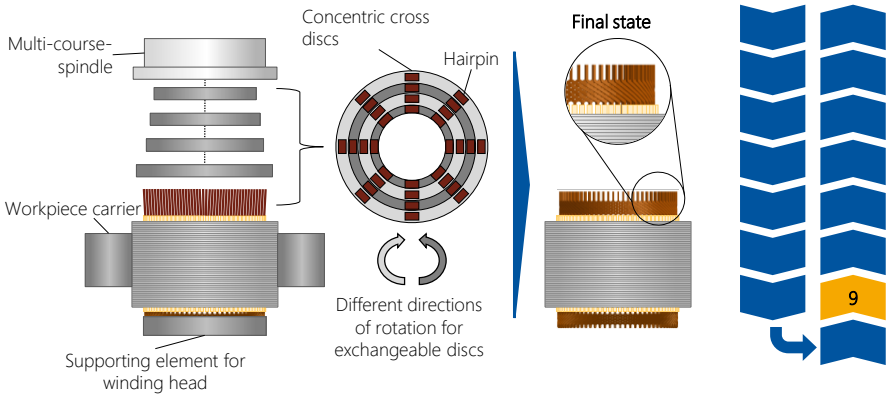
- Mechanical material parameters
- Initial orientation of the copper ends
- Wire cross section
- Surfaces of parts in contact with components

### Production costs\* [extract]

Machinery/equipment investment: € 0.08 – 0.12 million

# Twisting copper ends

Transverse arrangement according to the winding scheme



## Process description

- The protruding copper ends are formed tangentially in layers and brought into their final shape
- The copper ends are formed by a rotary tool movement (the tool has to be translatorily guided)
- The aim is to adjust the copper ends parallel to each other, which requires an S-shaped twisting geometry
- Free wire ends are supported in the bending radius during the twisting process
- If necessary, supporting elements are used on the opposite winding head to counteract the axial lifting force of the bending process

## Further information

### Technology alternatives [extract]

- Prefabricated wiring ring
- Continuous hairpin winding
- 3D-printed winding head

### Quality characteristics [extract]

- Geometric tolerances of the copper ends
- No damage to the conductor insulation
- No damage to the slot insulation
- Height tolerance

### Quality influences [extract]

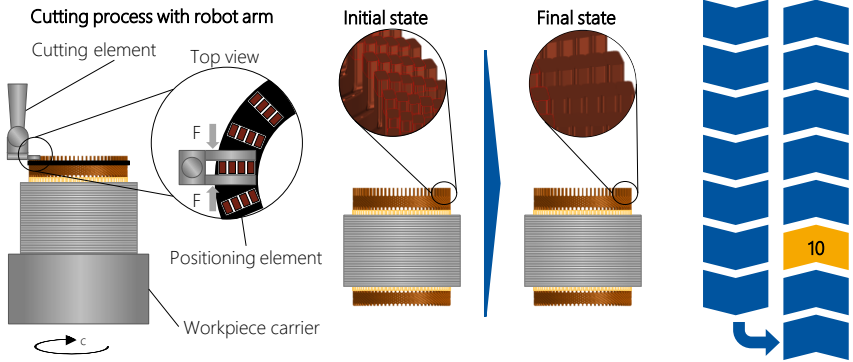
- Mechanical material parameters
- Initial orientation of the copper ends
- Wire cross section
- Surfaces of parts in contact with components

### Production costs\* [extract]

Machinery/equipment investment: € 0.8 – 2.5 million

# Cutting copper ends

Height adjustment for the welding process



## Process description

- To compensate for the differences in height of the copper ends to be contacted with each other, the copper ends are adjusted with a cutting device (shear cutting)
- To fix and protect the copper ends during the process, a positioning element is used, which is also partly used for repositioning
- There must be no copper sections remain in the winding head (danger of short circuit)
- The process can be carried out individually, slot by slot or in full cut, which has an influence on the achievable cycle time and variant flexibility

## Further information

### Technology alternatives [extract]

- Mechanical (grinding, cutting)
- Tolerance compensation system
- Machining
- 3D-printed winding head
- Wiring matrix
- Optional: This process step can be omitted if the axial tolerance is appropriately protected

### Quality characteristics [extract]

- No impurities in and on the wire surface
- Geometric tolerances
- No damage to the insulation
- No burr formation

### Quality influences [extract]

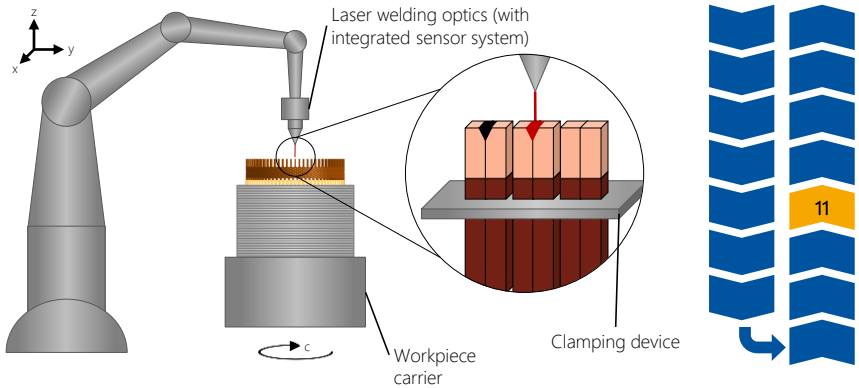
- Mechanical material parameters
- Tool wear
- Cutting speed
- Surfaces of parts in contact with components
- Complexity of the winding head

### Production costs\* [extract]

Machinery/equipment investment: € 0.15 – 1.0 million

# Contact copper ends

according to the specified winding scheme



## Process description

- The copper ends are electrically contacted with each other according to the specified winding scheme (usually substance-bound)
- Prior to the welding process, a camera-based sensor system with integrated control unit should also be used, which detects the position of the copper ends and automatically optimizes them
- Adjacent copper ends are fixed by a clamping device (theoretical zero gap formation)
- The surfaces of the copper ends are melted by absorption of the laser power and are substance-bounded connected with each other after the subsequent cooling process
- Depending on the winding head geometry, subsequent galvanic separation of adjacent welded connections may be necessary to prevent electrical short circuits

## Further information

### Technology alternatives [extract]

- Resistance soldering/resistance welding
- Electron beam welding
- Mechanical processes (e.g. hot crimping)

### Quality characteristics [extract]

- Low splash and pore formation
- Electrical conductivity of the weld seam
- Tensile strength of the weld seam
- Low thermal input
- Weld seam geometry (height of the welding point)
- Homogeneous weld cross section

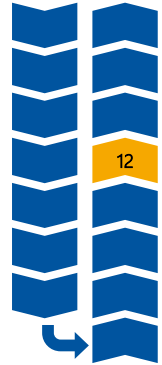
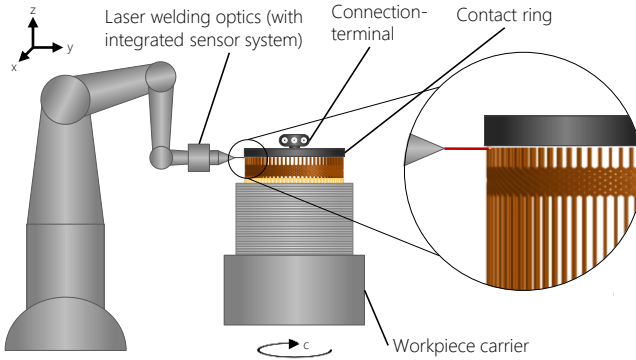
### Quality influences [extract]

- Energy input, laser power, wavelength
- Welding feed, focusing
- Welding preparation (stripping, surface cleaning etc.)
- Material selection (alloy content)
- Position of the copper ends (welding gap)

### Production costs\* [extract]

Machinery/equipment investment: € 0.4 – 0.8 million  
(per machine)

# Contact connecting elements with previous assembly



## Process description

- In this process step, the term “connecting elements” means the contact ring including jumpers/bridges, terminal connectors and star connectors
- The connecting elements are mounted in order to enable the contacting of two non-adjacent copper ends and thus the connection between winding sets
- After assembly and before the welding process, a camera-based sensor system can be used to correctly position the connecting elements
- By absorption of the laser power, the surfaces of the copper ends and the connecting elements are melted and, after the subsequent cooling process, are substance-bounded connected with each other

## Further information

### Technology alternatives [extract]

- Brazing
- Mechanical processes (e.g. hot crimping)

### Quality characteristics [extract]

- Low splash and pore formation
- Electrical conductivity of the weld seam
- Tensile strength of the weld seam
- Low thermal input
- Weld seam geometry (height of the welding point)
- Form and position tolerances of the interconnections

### Quality influences [extract]

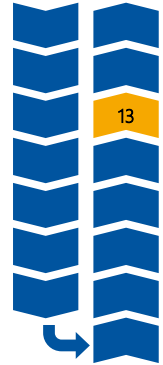
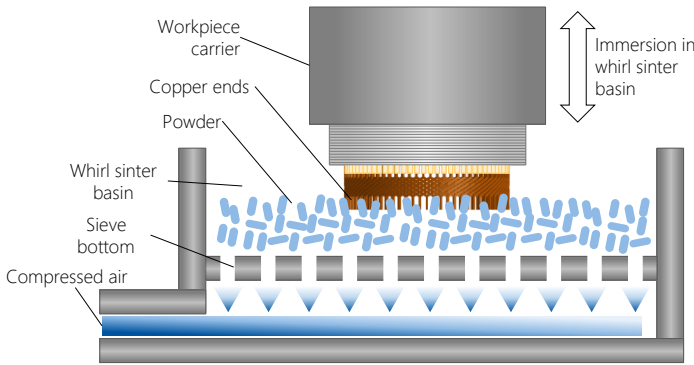
- Energy input, laser power, wavelength
- Welding feed, focusing
- Welding preparation (stripping, surface cleaning etc.)
- Position of the junctions (welding gap)

### Production costs\* [extract]

Machinery/equipment investment: € 0.4 – 0.8 million

# Insulating copper ends

for protection against short circuits and electric breakdown



## Process description

- When the resin in the stator has cured, the still bare welded copper ends have to be coated to prevent short circuits and leakage currents
- The whirl sintering process is often used, in which a powder coating is applied
- After preheating the copper ends, the copper ends are immersed in the sinter basin while turning
- The whirled powder melts on the copper surface as long as the surface temperature is above the melting temperature of the plastic
- The coating thickness depends on the preheating temperature and the immersion time
- After melting, cooling takes place on an outlet section

## Further information

### Technology alternatives [extract]

- Immersion method
- Trickling procedure
- Full potting

### Quality characteristics [extract]

- Coating thickness of the resin
- Dielectric strength
- No air inclusions in the insulation

### Quality influences [extract]

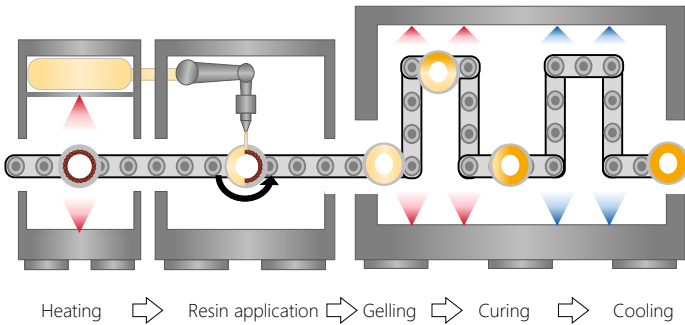
- Relative position of the copper ends to each other
- Gap between radial layers
- Homogeneous density of the particle size distribution in the whirl sinter basin
- Process temperature profile
- Application quantity of the resin

### Production costs\* [extract]

Machinery/equipment investment: € 0.1 – 1.0 million

# Stator impregnation

## Impregnation of the stator slot and winding head



## Process description

- Heating of the stator and resin before application
- Application of the impregnating resin by trickling procedure or immersion process on the winding heads
- Slot penetration by capillary effects
- Rotation of the stator for homogeneous resin distribution
- Precise gelling, curing and cooling of the resin by passing through specific temperature profiles or plant sections
- Use of full potting processes if higher heat dissipation of the stator is required
- Stator clamping concept important for freedom from damage in the variable process temperature profile

## Further information

### Technology alternatives [extract]

- Technology alternatives for resin application: trickling, roller dipping/rolling, hot dipping, vertical dipping, full potting
- Technology alternatives for heat input: convection, induction, electric heat (joule effect), infrared, ultraviolet radiation (UV radiation)
- Additional: Weight control before and after the process to determine the resin absorption and camera monitoring for application control

### Quality characteristics [extract]

- High fill factor (especially at the bottom of the slot)
- Avoidance of unfilled cavities
- High thermal conductivity
- Little reworking required for cleaning
- Aging resistance
- No damage to the laminated sheet metal package

### Quality influences [extract]

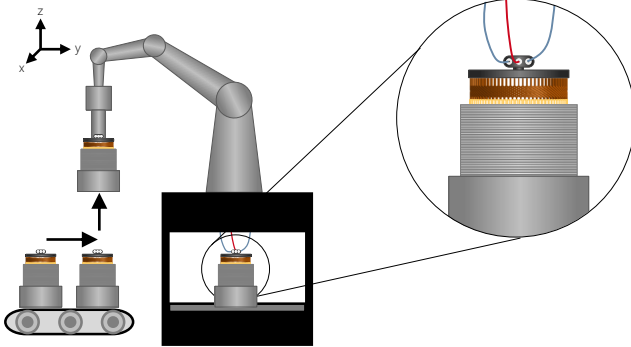
- Process temperature profile
- Application quantity of the resin
- Resin temperature profile
- Rotation speed
- Continuity/synchronisation of rotation
- Slot geometry

### Production costs\* [extract]

Machinery/equipment investment: € 0.2 – 5.0 million

# Stator test

Stator test at the end of stator production



## Process description

- Before the stator is assembled in further steps with other components to form the overall system, the stator is tested (usually this test is carried out at the end of the stator production line)
- During the stator test, the following things are tested amongst others:
  - Insulation strengths
    - Quality relevant: Internal insulation between windings within one or different phase(s)
    - Security relevant: External insulation between the conductor system and the laminated sheet metal package
  - Resistance: Testing the ohmic resistances within the system
  - Others: Polarization index test, step voltage test, rotational direction test

## Further information

### Possible testing technologies [excerpt]

- Surge voltage test and partial discharge measurement for testing the internal insulation; rule of thumb for the level of the test voltage:  $(2 \times U_{Nom} + 1,000V)$
- High voltage test (AC and/or DC) & partial discharge measurement to test the external insulation (dielectric strength); rule of thumb for the level of the test voltage:  $(2 \times U_{Nom} + 1,000V) \times (1.5...2)$

### Quality characteristics [extract]

- Symmetry of the phase resistances
- Insulation strength of winding/winding and winding/housing
- Freedom from partial discharge of the entire system
- Error in the winding scheme

### Quality influences [extract]

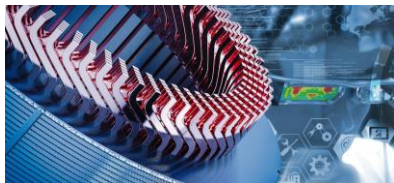
- Insufficient contacting
- Test parameter setting

### Production costs\* [extract]

Machinery/equipment investment: € 0.05 – 0.6 million



The PEM of RWTH Aachen University has been active in the field of electric motor production for many years. The research division Electric Drive Production is concentrating in particular on hairpin technology with a focus on the necessary production facilities. By conducting numerous national and international industrial projects with companies on all levels of the value chain as well as central positions in renowned research projects, PEM offers extensive expertise in this area.



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With more than 3,300 members, the VDMA is the largest network organisation of the mechanical and plant engineering industry in Germany and Europe. The Power Transmission Engineering Association supports its members on technological, economic and political issues and provides information on current topics and trends in drive technology and in mechanical and plant engineering. The VDMA forum "E-MOTIVE" deals with technical aspects and applications of electromobility.



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Do you have any questions?

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